DAY 17 | POST-READ

DevOps and Cloud: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
DevOps	Culture and practices: shared ownership, small changes, automation, blameless learning.
DevOps loop	Plan, code, build and test, release, operate and monitor, repeat.
CI	Every push is automatically built and tested.
Continuous delivery	Every green build is release-ready; a person approves.
Continuous deployment	Every green build goes to production automatically.
IaaS / PaaS / SaaS	Less to manage and less control as you move right.
Azure hierarchy	Tenant, subscription, resource group, resources.
Azure DevOps	Boards, Repos, Pipelines, Test Plans, Artifacts.
Pipeline YAML	trigger, pool, steps (or stages and jobs) in azure-pipelines.yml.

Part B: Going beyond today

1. The shared responsibility model

Security is shared. In every model the cloud provider secures the physical data centres; you always remain responsible for your data, your users and how you grant access.

2. DORA metrics

Teams often track four delivery measures: deployment frequency, lead time for changes, change failure rate and time to restore service. They show whether DevOps practices are actually working.

3. Infrastructure as Code

```
az deployment group create --resource-group rg-demo --template-file main.bicep
```

Instead of clicking in the portal, describe resources in files (Bicep, ARM or Terraform), review them like code and recreate environments identically.

4. Tags and cost control

```
az group create -n rg-demo -l centralindia --tags owner=ahmed project=training
```

Tag resources with owner and project, set budgets and alerts, and delete test resource groups when finished. Check current pricing and free offers on the Azure website, as they change.

5. GitHub Actions

```
on: [push]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: pip install -r requirements.txt && pytest -q
```

The same CI ideas with different syntax. Knowing one makes the other easy.

Part C: Check yourself

- What is the difference between continuous delivery and continuous deployment?
- Give one example each of IaaS, PaaS and SaaS on Azure.
- Why are small, frequent releases safer?
- What happens when you delete a resource group?
- Which Azure DevOps service would you use for pull requests?

Part D: Practice ideas for tonight

- Sign in to the Azure portal and open Cloud Shell. Run az account show (do not create paid resources yet).
- Write an azure-pipelines.yml for your Day 13 tests.
- Classify ten tools you use at work as IaaS, PaaS or SaaS.

Coming up on Day 18: Git and Azure Repos, Azure Pipelines YAML for build and test automation, and Docker images.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today